

Nigerian Electricity Fraud Detection Using Smart Meter Data

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Focus Area: Energy

1.0 Project Overview

Electricity theft causes major revenue losses for Nigerian power companies. This project uses machine learning to analyze smart meter data and detect possible fraud automatically. Using 15,000 meter records from 500 customers across five DisCos, the model achieved 99.97% accuracy. The solution enables DisCos to move from reactive, inspection-based investigations to proactive, data-driven fraud detection, improving revenue protection and operational efficiency.

2.0 Problem Statement

Electricity theft through meter tampering and illegal connections causes significant revenue losses and puts extra pressure on the power grid causing outages, equipment failures, and higher maintenance costs.. Most DisCos still rely on customer complaints and manual inspections to detect fraud, which is slow, expensive, and often ineffective. This project aims to develop a data-driven system that can automatically identify suspicious electricity usage and help detect fraud before major losses occur.

3.0 Dataset Overview

- **Total Data:** 15,000 smart meter readings
- **Final Clean Dataset:** 14,849 rows × 15 columns
- **Time Period:** 30 days
 - **DisCos Covered:** Abuja AEDC, Kano KEDCO, Eko DisCo, Ikeja Electric, Jos JED
- **Target Variable:** `Target_Anomaly` (0 = Normal, 1 = Fraud)

Class Distribution: Normal meters had 91.41% **and** Tampered meters had 8.59%

Dataset imbalance was handled using: `class_weight='balanced'` in Random Forest

4.0 Data Cleaning Summary

Key preprocessing steps:

- Removed missing DisCo records (151 rows)
- Replaced sensor error codes (-99) with NaN
- Imputed missing values using **tariff-band median**
- Retained voltage outliers as they represent fraud signals
- Final dataset had **no missing values**

5.0 Feature Engineering

New features were created to help the model detect unusual electricity usage, such as sudden drops in consumption(**Consumption_Drop_From_Mean**), voltage changes(**Voltage_7d_std**), leakage patterns(**Leakage_7d_max**), and recent usage trends(**Consumption_Lag_1**). These features improved fraud detection while ensuring no data leakage during model training.

6.0 Key Insights (EDA Findings)

The analysis showed that Abuja AEDC (**12.34%**) had the highest fraud rate, while Jos JED (**6.02%**) had the lowest. Customers in Tariff Bands C (**12.01%**) and D (**11.04%**) were more likely to tamper with meters. The strongest fraud indicators were high current leakage (+0.68), low power factor(-0.60), lower voltage (-0.51), and reduced electricity consumption(-0.34). Customer complaints showed little connection to actual fraud cases

7.0 Statistical Analysis

Statistical testing confirmed that the key fraud indicators were highly significant. Factors such as consumption drops, leakage current spikes, and tariff band categories showed a strong

relationship with fraud, all had a p-value of $p < 0.001$ indicating that the detected patterns were genuine and not due to chance.

8.0 Machine Learning Model

A **Random Forest Classifier** was used to detect electricity fraud because it handles complex patterns, noisy sensor data, and imbalanced datasets effectively.

Model Performance

- **Accuracy:** 99.97%
- **Precision:** ~1.00
- **Recall:** 1.00
- **F1-Score:** ~1.00
- **ROC-AUC:** 1.00

The model achieved near-perfect performance on the test set, successfully identifying fraudulent meters while minimizing false alarms. To ensure validity, the data was split before feature engineering, preventing data leakage and ensuring the model learned only from historical information.

Top Predictive Features

1. Leakage_7d_max (7-day maximum leakage)
2. Peak_Demand_kW
3. Consumption_7d_mean
4. Current_Leakage_Amp
5. Avg_Voltage

These results show that smart meter telemetry can be highly effective for detecting electricity fraud and supporting proactive revenue protection effort

9.0 Tableau Dashboard

An interactive Tableau dashboard was developed to support fraud monitoring and decision-making. The dashboard tracks key metrics such as 14,849 monitored meters, 1,275 fraud cases (8.6%), and an estimated ₦1.78 million revenue loss. It also highlights fraud hotspots, with Abuja AEDC and Kano KEDCO recording the highest losses, and identifies major risk indicators such as tampering alerts and high leakage current.

10.0 Recommendations

1. **Focus on High-Risk Regions** : Abuja AEDC and Kano KEDCO require priority monitoring
2. **Target Vulnerable Tariff Bands**: Band C and Band D customers show highest fraud probability
3. **Deploy Smart Meter Security Upgrades**: Use tamper-proof and alert-enabled meters
4. **Adopt Data-Driven Field Inspections**: Replace complaint-based systems with ML risk scoring

11.0 Conclusion

This project demonstrates how machine learning can help Nigerian DisCos detect electricity fraud more effectively using smart meter data. By identifying suspicious usage patterns early, the system can reduce revenue losses, improve grid reliability, and support more efficient field operations. The approach enables a shift from reactive investigations to proactive, data-driven fraud monitoring. field operations

Tech Stack

- Python (Pandas, NumPy)
- Scikit-learn (Random Forest)
- SciPy (Statistical testing)
- Matplotlib / Seaborn (EDA)
- Tableau Public (Dashboarding)
- Jupyter Notebook